

TrES-2b

R-1366

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_130614 · created 2026-08-01T00:10:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456457.89662 +/- 0.00023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.14 +/- 0.011
Transit depth (Rp/Rs)^2	1.95 +/- 0.32 [%]
Orbital Inclination (inc)	83.87 +/- 0.28
Airmass coefficient 1 (a1)	1.16 +/- 0.011
Airmass coefficient 2 (a2)	-0.135 +/- 0.041
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#2 - [151, 117]
Optimal Aperture	4.29
Optimal Annulus	12.2
Transit Duration (day)	0.0764 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.14 $\pm$ 0.011 vs 0.1254 (deviation 0.81 σ)
Duration fit vs published	0.0764 d vs 0.0746 d (deviation 0.25 σ)
Mid-time O–C	0.5 min
Depth SNR	7.8
Residual scatter	0.99 %

Light curve

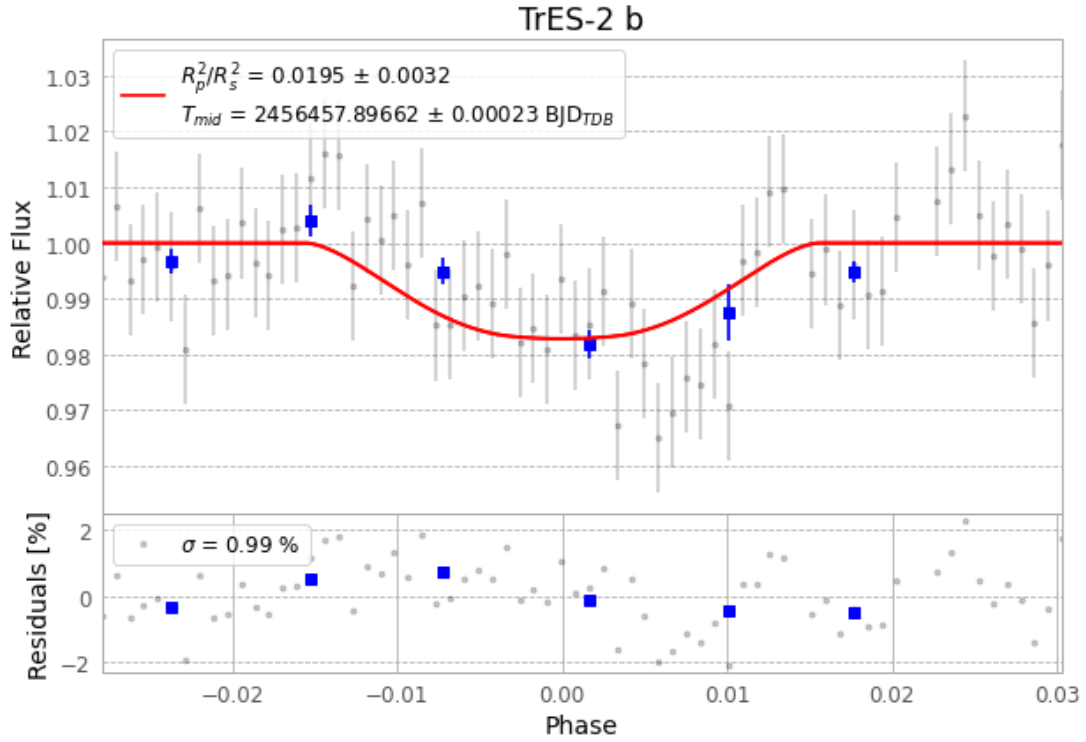


Figure 1 — FinalLightCurve_TrES-2 b_2013-06-14.png (SHA-256 0a78b68b388c738c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [424, 245], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f30ff87a158d455c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), TRES-2130614074812.FITS → TRES-2130614111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-130614.log (b3254384bef5...), FinalLightCurve_TrES-2 b_2013-06-14.png (0a78b68b388c...), FinalLightCurve_TrES-2 b_2013-06-14.csv (81b46d6be8b6...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 00:10Z.